

Assembling selected GNS cyclic subspaces into an isometric source representation

MathlibAnnex.CStarAlgebra.CAR.exists_selectedAtomicCyclicIsometry

theorem

Joins mutually orthogonal pure-state cyclic copies and controls the limiting flag projections on their sum.

Statement

For the shell family and selected GNS spaces below, let $\sigma : A \rightarrow B(K)$ be a unital complex star representation on a complete complex Hilbert space. Suppose unit vectors $\eta_j \in K$ satisfy $\langle \eta_j, \sigma(a)\eta_j \rangle = \varphi_j(a)$ for every $j \in I$ and $a \in A$. Put $M_j = \overline{\{\sigma(a)\eta_j : a \in A\}}$ and let P_j be the orthogonal projection onto $\bigcap_n \text{ran } \sigma(f_{j,n})$.

There is a complex-linear isometry $W : \mathcal{H} \rightarrow K$ such that $W(e_j H_j) \subseteq M_j$, $W(e_j \xi_j) = \eta_j$ and $P_j Wx \in \mathbb{C}\eta_j$ for every j and $x \in \mathcal{H}$. For every $a \in A$, it satisfies $W\pi(a) = \sigma(a)W$.

Assumptions

Let A be the completed CAR algebra, the completion of the matrix stages $M_{2^n}(\mathbb{C})$ under $a \mapsto a \otimes I_2$. Let f_n be the image of the first diagonal matrix unit, so $f_0 = 1$ and f_n decreases through projections. The root state ω takes the value a_{00} on each matrix stage.

Choose one pure state φ_j from each unitary-equivalence class $j \in I$ of pure-state Gelfand-Naimark-Segal (GNS) representations, with root index o and $\varphi_o = \omega$. Write (H_j, π_j, ξ_j) for its complete complex GNS space, unital star representation and unit cyclic vector. Set $\mathcal{H} = \bigoplus_{j \in I} H_j$, $\pi = \bigoplus_j \pi_j$, and let $e_j : H_j \rightarrow \mathcal{H}$ be the coordinate embedding. No countability of I is assumed. Inner products are linear in the second argument.

A supplied shell family consists of unital complex star automorphisms α_j and elements $w_{j,n} \in A$. Put $f_{j,n} = \alpha_j(f_n)$. The identities are $\varphi_j \circ \alpha_j = \omega$, $w_{j,n}^* w_{j,n} = f_{j,n} - f_{j,n+1}$ and $w_{j,n} w_{j,n}^* = f_n - f_{n+1}$. At the root, α_o is the identity and $w_{o,n} = f_n - f_{n+1}$. The family is an input, not an existence conclusion. The vectors η_j are given simultaneously, one for each chosen class. No irreducibility of σ and no density of any individual orbit in all of K is assumed. The algebra and the representations are unital, so each η_j belongs to M_j .

Conclusion

The isometry identifies the atomic Hilbert sum with a closed subspace reducing σ . Each ambient limiting flag projection sends this range into the indicated cyclic line. This is control on the isometry range; it is not a claim that the entire ambient fixed space has dimension one.

Surjectivity is not part of this theorem. The surjective cyclic-sum theorem proves it in the generated-target setting with additional hypotheses. The present hypotheses already supply the vector states; they do not assert that every representation of the CAR algebra contains all selected classes.

Proof route

Identify each cyclic restriction with its selected GNS representation by pointed cyclic transport. Orthogonality of inequivalent cyclic pieces allows these identifications to extend to the arbitrary-index Hilbert sum. The own-piece compression identity and vanishing on other cyclic pieces give the projection assertion; continuity passes the source intertwining law through the convergent sum.

▼ Proof steps

1. On M_j , the restricted representation has the unit cyclic vector η_j with pure state φ_j , and is irreducible. Equality of vector states gives a unitary $U_j : H_j \rightarrow M_j$ sending ξ_j to η_j and intertwining π_j with the restriction of σ . It extends the orbit map $\pi_j(a)\xi_j \mapsto \sigma(a)\eta_j$.
2. For $i \neq j$, the orthogonal projection from M_j to M_i intertwines their restrictions, since the cyclic subspaces reduce σ . A nonzero such intertwiner between irreducible representations would give a unitary equivalence by Schur's lemma, contradicting the distinct selected GNS classes. Thus $M_i \perp M_j$. The maps U_j , followed by inclusion into K , form an orthogonal family of isometries.
3. For a fixed i , the state identity gives $\varphi_i(f_{i,n}) = 1$, hence $\sigma(f_{i,n})\eta_i = \eta_i$. Norm compression implies $P_i\sigma(a)P_i = \varphi_i(a)P_i$, so $P_i\sigma(a)\eta_i = \varphi_i(a)\eta_i$. Density of the orbit and closedness of the line give $P_iM_i \subseteq \mathbb{C}\eta_i$. If $x \in M_j$ with $j \neq i$ and $P_ix \neq 0$, normalizing P_ix yields a vector with state φ_i . Its cyclic space is orthogonal to M_j . Then $\langle P_ix, x \rangle = \|P_ix\|^2 = 0$, a contradiction. Hence $P_iM_j = 0$.
4. For $x = (x_j) \in \mathcal{H}$, define $Wx = \sum_j U_j x_j$, with each summand viewed in K . Orthogonality makes this an unconditionally norm-convergent Hilbert sum and gives $\|Wx\|^2 = \sum_j \|x_j\|^2$. The coordinate formulas give $W(e_j\xi_j) = \eta_j$. Applying the bounded operator P_i kills every summand except the i -th, whose image lies in $\mathbb{C}\eta_i$. Applying the bounded operator $\sigma(a)$ termwise gives $W\pi(a) = \sigma(a)W$.

Main citations

- Exact theorem and proof
- The supplied shell-family data
- The chosen pure-GNS Hilbert sum
- Pointed unitary identification of one cyclic piece
- Schur orthogonality for different selected classes
- Compression sends its own cyclic piece into one line
- The limiting projection vanishes on other cyclic pieces
- Additional hypotheses for surjectivity
- The root state with its purity proof

Lean source signature (exact)

```
theorem exists_selectedAtomicCyclicIsometry
  (family : RepresentativeShellFamily)
  {K : Type v} [NormedAddCommGroup K] [InnerProductSpace ℂ K]
  [CompleteSpace K] (sigma : Representation Limit K)
  (eta : MathlibAnnex.CStarAlgebra.PureState.GNSClass Limit → K)
  (heta : ∀ i, ‖eta i‖ = 1)
  (hstate : ∀ i, Representation.vectorFunctional sigma (eta i) =
    (MathlibAnnex.CStarAlgebra.PureState.representative completedRootPureState i).1) :
```

```

    ∃ W : MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert completedRootPureState
→i[C] K,
    (∀ i x, W (MathlibAnnex.CStarAlgebra.PureState.selectedEmbedding
completedRootPureState i x) ∈
    Representation.cyclicSubspace sigma (eta i)) ∧
    (∀ i, W (MathlibAnnex.CStarAlgebra.PureState.selectedEmbedding completedRootPureState
i
    (MathlibAnnex.CStarAlgebra.PureState.selectedVector completedRootPureState i)) =
eta i) ∧
    (∀ i x, commonFixedProjection
    (fun n ↦ sigma (transportedFlag family i n)) (W x) ∈
    C · eta i) ∧
    ∀ a,
    W.toContinuousLinearMap.comp
    (selectedAtomicRepresentation a) =
    (sigma a).comp
    W.toContinuousLinearMap

```

In the source	Mathematical meaning
family : RepresentativeShellFamily	The given CAR shell family with transported flags $f_{j,n} = \alpha_j(f_n)$ and selected pure-GNS data (H_j, π_j, ξ_j) , indexed by arbitrary I . Write $\mathcal{H} = \bigoplus_j H_j$, $\pi = \bigoplus_j \pi_j$.
sigma : Representation Limit K; [CompleteSpace K]	The given unital star representation $\sigma : A \rightarrow B(K)$ on a complete complex Hilbert space; it need not be irreducible.
eta : GNSClass Limit → K; heta : ∀ i, eta i = 1	A simultaneously given family of unit vectors $\eta_j \in K$, one for each selected class. Source i denotes the mathematical index j .
hstate : ∀ i, Representation.vectorFunctional sigma (eta i) = (PureState.representative completedRootPureState i).1	For every j and every $a \in A$, $\langle \eta_j, \sigma(a)\eta_j \rangle = \varphi_j(a)$. Equality is of whole functionals; it is not just one matrix coefficient. Inner products are linear in the second entry.
∃ W : SelectedAtomicHilbert completedRootPureState → _i [C] K	One complex-linear isometry $W : \mathcal{H} \rightarrow K$ satisfies all four following properties. The arrow does not assert surjectivity.
∀ i x, W (selectedEmbedding ... i x) ∈ Representation.cyclicSubspace sigma (eta i)	For every j and $x \in H_j$, $W(e_j x) \in M_j = \overline{\{\sigma(a)\eta_j : a \in A\}}$. The orbit is a linear subspace because A and the representation are unital and linear. M_j is not assumed to be all of K .
∀ i, W (selectedEmbedding ... i (selectedVector ... i)) = eta i	The same W sends each embedded unit cyclic vector to its supplied vector: $W(e_j \xi_j) = \eta_j$.
∀ i x, commonFixedProjection (fun n ↦ sigma (transportedFlag family i n)) (W x) ∈ C · eta i	For every j and $x \in \mathcal{H}$, $P_j W x \in \mathbb{C} \eta_j$, where P_j projects onto $\bigcap_n \text{ran } \sigma(f_{j,n})$. This controls the projection only on the range of W , not the dimension of the entire ambient fixed space.

In the source	Mathematical meaning
$\forall a, W.\text{toContinuousLinearMap.comp}$ $(\text{selectedAtomicRepresentation } a) = (\text{sigma}$ $a).\text{comp } W.\text{toContinuousLinearMap}$	For every $a \in A$, $W\pi(a) = \sigma(a)W$. $W.\text{toContinuousLinearMap}$ is the same isometry viewed as a bounded map, and composition acts rightmost first.

Further source notes: Here Limit is A , $\text{completedRootPureState}$ is the root state ω bundled with its purity proof, and $\text{SelectedAtomicHilbert } \text{completedRootPureState}$ is $\mathcal{H}$. The arrow $\rightarrow_{i1}[\mathbb{C}]$ asserts a complex-linear isometry; it does not assert surjectivity.

Earlier published Card and PDF

▼ Exact content identity

Declaration: `MathlibAnnex.CStarAlgebra.CAR.exists_selectedAtomicCyclicIsometry`

Accepted content SHA-256: `c0bfb055058f001ab432d17e0bc4b3ab89842a3c706cc3c84461900b266c4de2`

Accepted source guide SHA-256: `1c48447e5b17bc9705fe5825ec82ab86d1e4d2f976caaa5c5493095357f80b69`

Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`